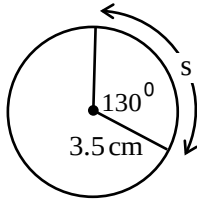


Question 1

2 marks

101

Use the information in the diagram to determine the value of the arc length “s”.



Question 2

2 marks

102

Solve the following equation over the interval $[0, 2\pi)$.

$$\tan^2 \theta + 2.8 \tan \theta + 1.96 = 0$$

Use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$.

Question 3

3 marks

103

Determine how many monthly investments of \$50 would have to be deposited into a savings account that pays 3% annual interest, compounded monthly, for the account's future value to be \$50,000.

Use the formula:

$$FV = \frac{R \left[(1 + i)^n - 1 \right]}{i}$$

where FV = the future value

R = the investment amount

$i = \frac{\text{the annual interest rate}}{\text{the number of compounding periods per year}}$

$n = \text{the number of investments}$

Express your answer as a whole number.

Question 4

2 marks

104

There are 5 men and 4 women to be seated in a row.

How many arrangements are possible if two men must sit at the beginning of the row and two men must sit at the end of the row?

Question 5

a) 3 marks b) 2 marks

105
106

a) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^8$, determine the 3rd term.

b) In the binomial expansion of $\left(\frac{3}{x^2} - 4x^5\right)^n$, the 6th term contains x^{25} .
Solve for n .

Note: A calculator is not required for the remaining test questions.

Question 6**1 mark**

107

Given the following two functions, $f(x) = \sqrt{x-1}$ and $g(x) = x^2 + 1$, evaluate $g(f(3))$.

Question 7**2 marks**

108

If θ terminates in quadrant II and $\csc \theta = \frac{3}{2}$, determine the exact value of $\tan \theta$.

Question 8

a) 1 mark b) 1 mark

109
110

a) Determine the remainder when $x^4 - 3x^2 + 1$ is divided by $x + 2$.

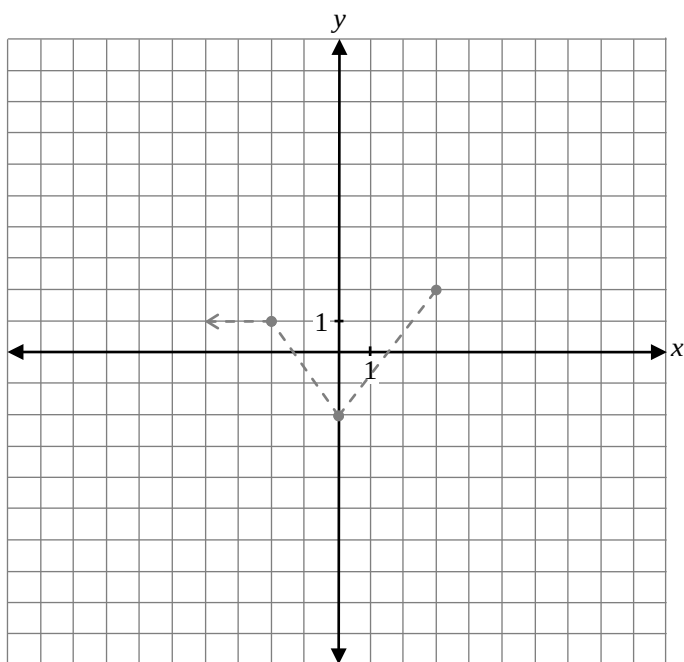
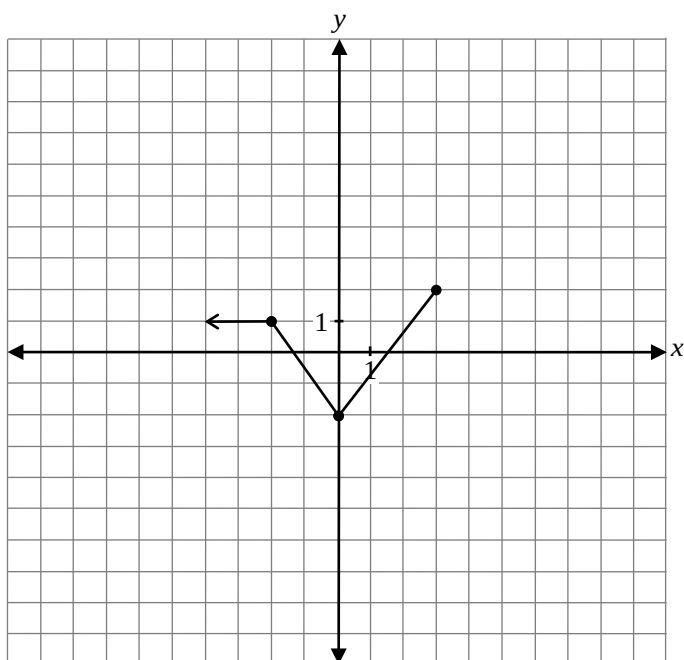
b) Is $x + 2$ a factor of $x^4 - 3x^2 + 1$?
Explain your reasoning.

Question 9

2 marks

111

Given the graph of $y = f(x)$ below, sketch the graph of $y = 2f(x) - 3$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 10**1 mark**

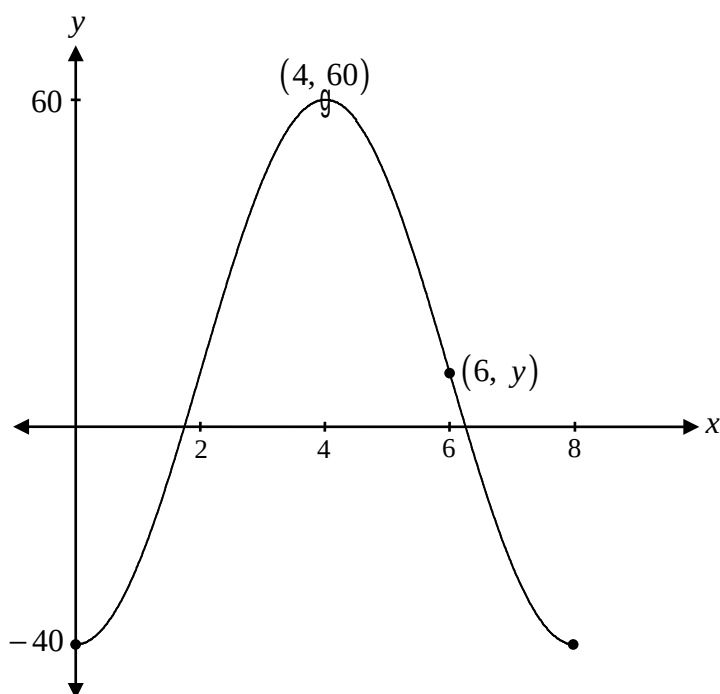
112

Determine one possible restriction for the domain of $f(x) = (x-1)^2$ so that the inverse of $f(x)$ is a function.

Question 11**1 mark**

113

Using the graph of the sinusoidal function below, find the value of y in the point $(6, y)$.



Question 12

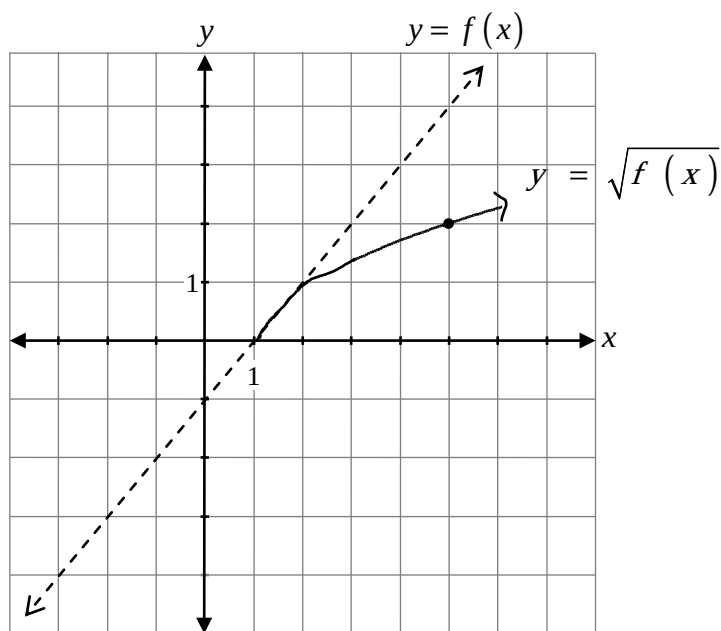
1 mark

114

Billy was given the graph of $y = f(x)$.

He was asked to sketch the graph of $y = \sqrt{f(x)}$.

His answer is given on the graph below.



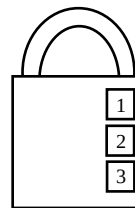
Explain the error Billy made when sketching the graph of $y = \sqrt{f(x)}$.

Question 13

1 mark

115

Explain why a locker combination should really be called a locker permutation.



Question 14

1 mark

116

The graph of $f(x) = x^2 + 4$ is reflected over the x -axis.

Write the equation of the new function.

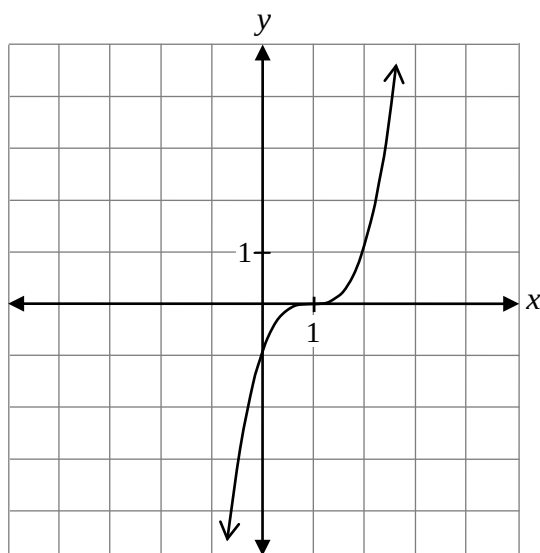
$y =$ _____

Question 15

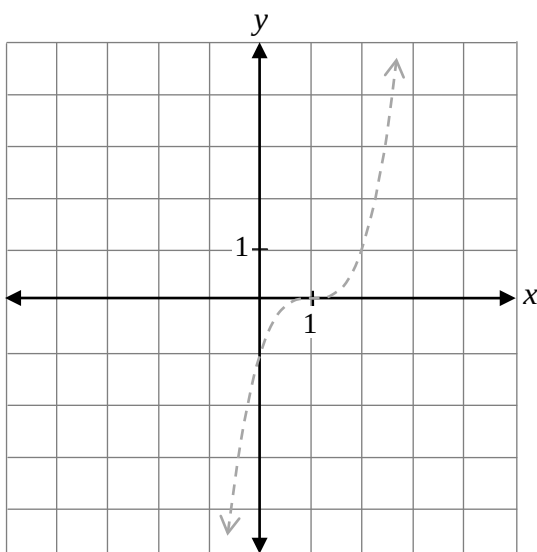
2 marks

117

Given the graph of $y = f(x)$ below,



sketch the graph of $y = \frac{1}{f(x)}$.



The graph of $f(x)$ has already been drawn for your reference.

No marks will be awarded for the graph of $f(x)$.

Question 16

2 marks

118

Divide $(x^3 - 5x - 4)$ by $(x + 1)$.

Question 17

1 mark

119

You are given the following row of Pascal's Triangle.

1 7 21 35 35 21 7 1

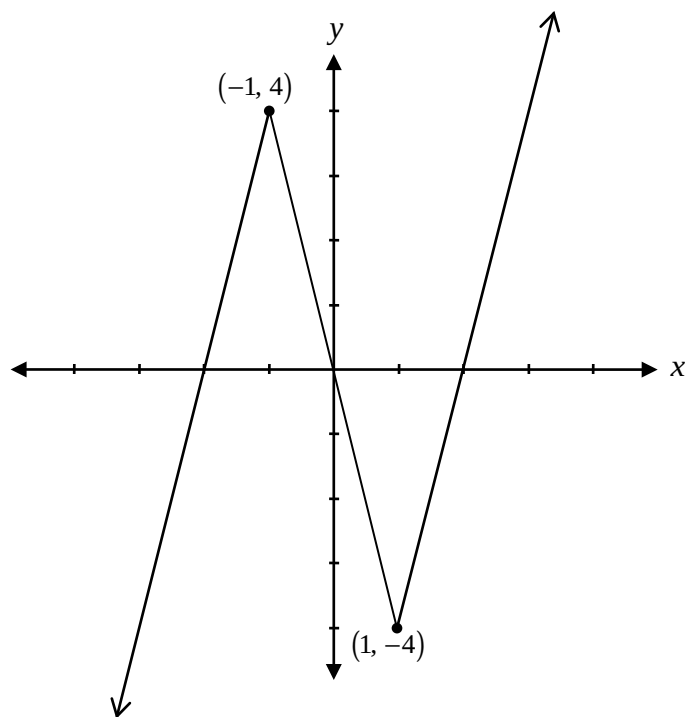
Determine the values of the next row.

Question 18

2 marks

120

Given the graph of $y = f(x)$ below, state the domain and range of $y = \sqrt{f(x)}$.



Domain: _____

Range: _____

Question 19

3 marks

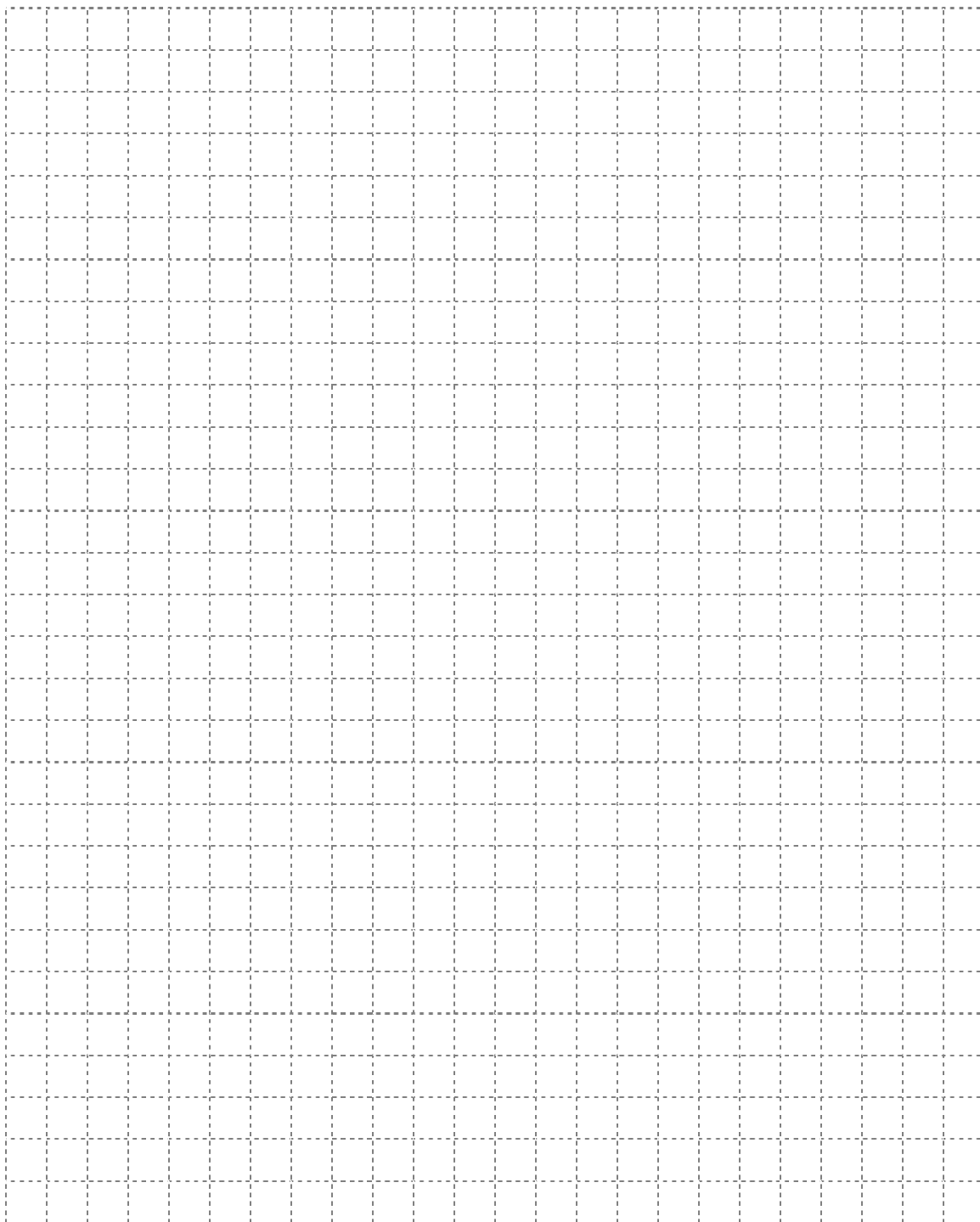
121

Prove the identity below for all permissible values of θ :

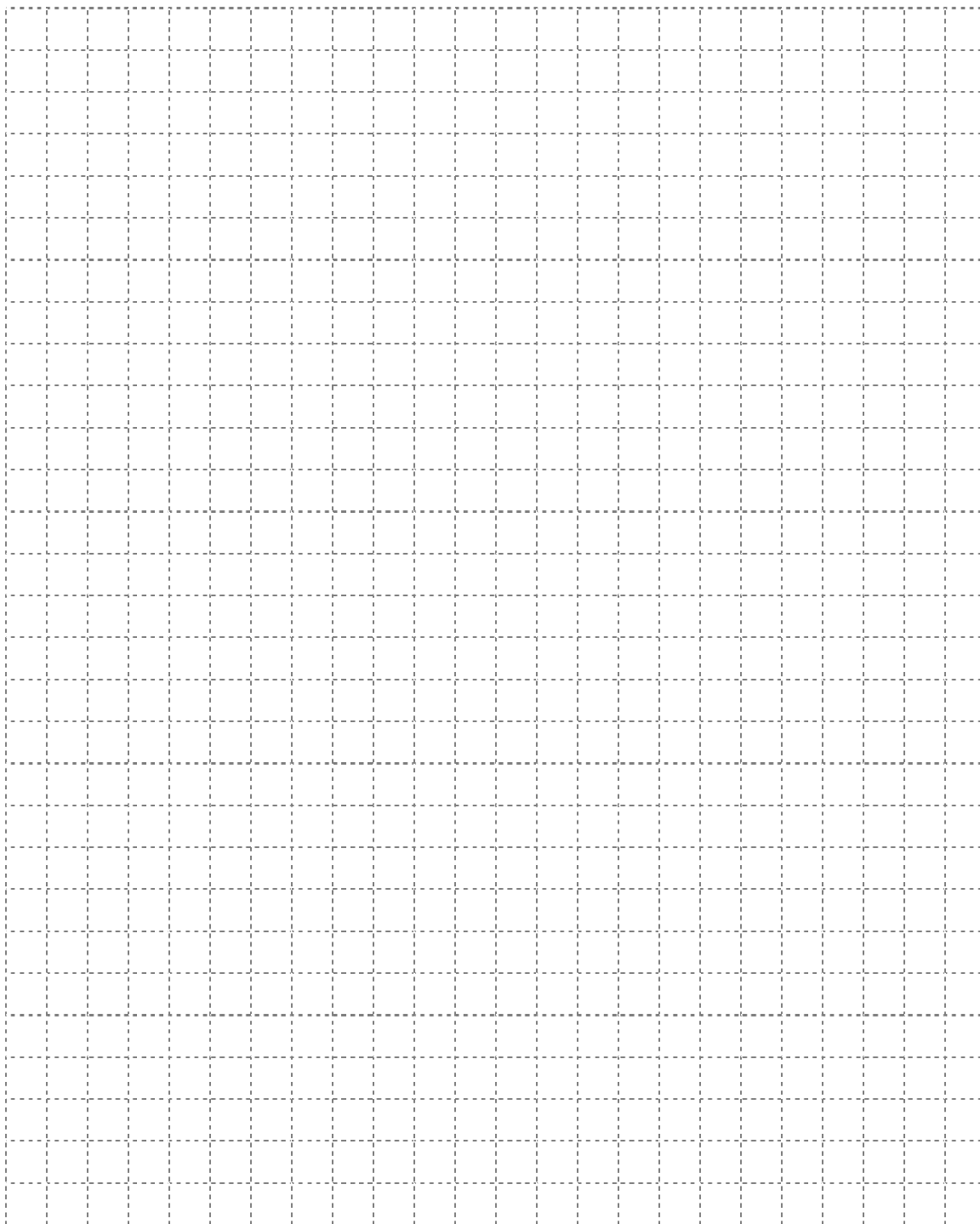
$$\frac{1 - \tan^2 \theta}{1 + \tan^2 \theta} = \cos 2\theta$$

Left-Hand Side	Right-Hand Side

No marks will be awarded for work done on this page.



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